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Genome Note

Whole-genome sequence of a putative pathogenic *Bacillus* sp. strain SD-4 isolated from cattle feed

Sajid Iqbal[a], Muhammad Faraz Bhatti[b], Aneela Javed[c], Kashif Rahim[d], Rehan Zafar Paracha[e], Hussnain Ahmed Janjua[a,*]

[a] *Department of Industrial Biotechnology, Atta-ur-Raman School of Applied Biosciences (ASAB), National University of Sciences and Technology (NUST), Islamabad, Pakistan*
[b] *Department of Plant Biotechnology, Atta-ur-Raman School of Applied Biosciences (ASAB), National University of Sciences and Technology (NUST), Islamabad, Pakistan*
[c] *Department of Healthcare Biotechnology, Atta-ur-Raman School of Applied Biosciences (ASAB), National University of Sciences and Technology (NUST), Islamabad, Pakistan*
[d] *Department of Microbiology, Cholistan University of Veterinary and Animal Sciences (CUVAS), Bahawalpur, Pakistan*
[e] *Research Centre for Modelling and Simulation (RCMS), National University of Sciences and Technology (NUST), Islamabad, Pakistan*

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ABSTRACT

Objectives: The present study describes the draft genome sequence of a novel *Bacillus* sp. strain SD-4 isolated from animal feed. The study aims to get a deeper insight into antimicrobial resistance and secondary metabolite biosynthetic gene clusters (BGCs) and the association between them.
Methods: The strain SD-4 was preliminarily evaluated for antibacterial activities, motility, biofilm formation, and enterotoxin production using in vitro assays. The genome of strain SD-4 was sequenced using the Illumina HiSeq 2500 platform with paired-end reads. The reads were assembled and annotated using SPAdes and PGAP, respectively. The genome was further analysed using several bioinformatics tools, including TYGS, AntiSMASH, RAST, PlasmidFinder, VFDB, VirulenceFinder, CARD, PathogenFinder, MobileElement finder, IslandViewer, and CRISPRFinder.
Results: In vitro assays showed that the strain is motile, synthesises biofilm, and produces an enterotoxin and antibacterial metabolites. The genome analysis revealed that the strain SD-4 carries antimicrobial resistance genes (ARGs), virulence factors, and beneficial secondary metabolite BGCs. Further genome analysis showed interesting genome architectures containing several mobile genetic elements, including two plasmid replicons (repUS22 and rep20), five prophages, and at least four genomic islands (GIs), including one *Listeria* pathogenicity island LIPI-1. Moreover, the strain SD-4 is identified as a putative human pathogen.
Conclusion: The genome of strain SD-4 harbours several BGCs coding for biologically active metabolites. It also contains antimicrobial resistance genes and is identified as a potential human pathogen. These results can be used to better comprehend antibiotic resistance in environmental bacteria that are not influenced by human intervention.

1. Introduction

The *Bacillus cereus* group comprises at least 21 related species ranging from probiotics to pathogenic [1]. These bacterial species are ubiquitously distributed worldwide in diverse environments and are frequently isolated from water, soil, rhizosphere samples, and the intestinal tract of animals. As opportunistic pathogens, they are becoming one of the prominent causes of food poisoning [2]. *B. cereus* can cause two types of gastrointestinal disease;

* Corresponding author. Department of Industrial Biotechnology, Atta-ur-Raman School of Applied Biosciences (ASAB), National University of Sciences and Technology (NUST), 44000 Islamabad, Pakistan\

E-mail addresses: hussnain.janjua@asab.nust.edu.pk, janjua.hussnain@gmail.com (H.A. Janjua).

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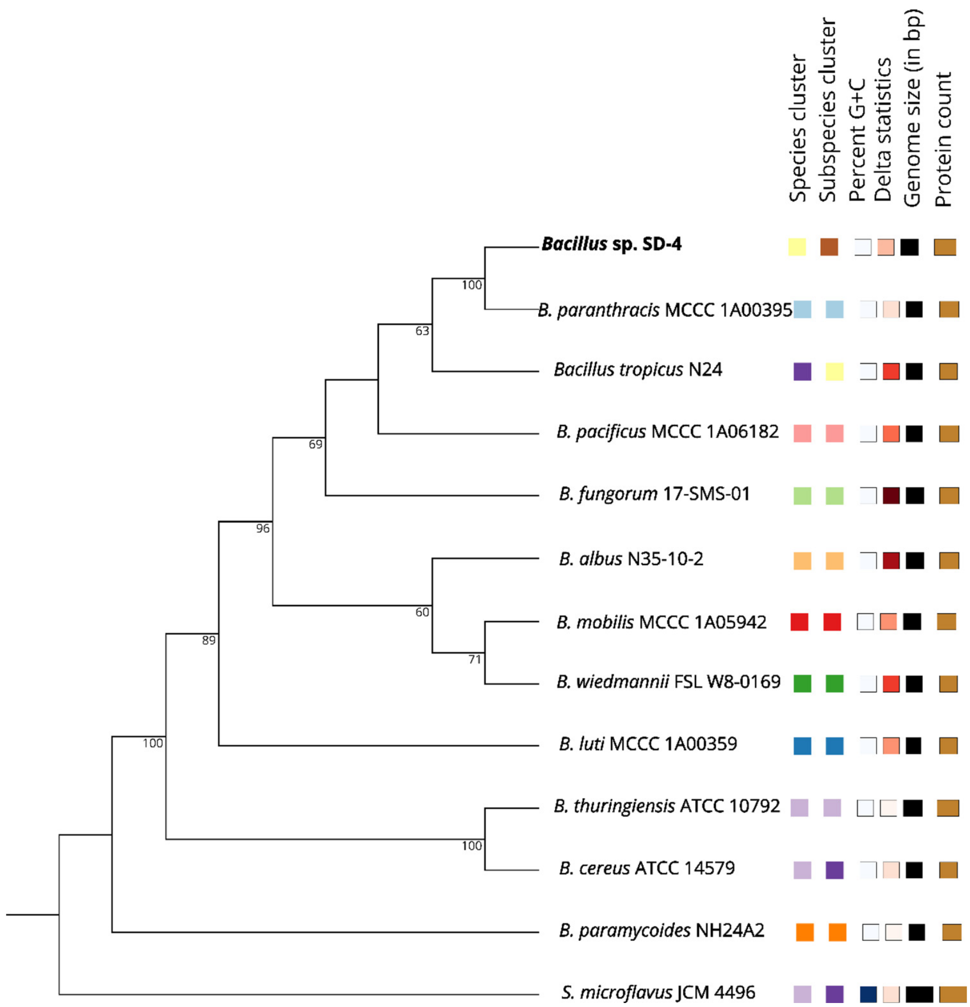

Fig. 1. The phylogenomic tree was constructed based on whole-genome sequences of *Bacillus* sp. strain SD-4 and closely related members of the genus *Bacillus* inferred using the MASH algorithm. Based on the smallest MASH distance, 10 closely related type strains were selected from TYGS database, and a balanced evolution tree with branch support was constructed via FASTME version 2.1.6.1. Bootstrap support values greater than 60% are shown at their respective nodes. *Streptomyces microflavus* strain JCM 4496 was used as the outgroup.

emetic and diarrheal syndrome. The non-hemolytic enterotoxin (*nhe*), cytotoxin K (*cytK*), and other virulence factors, such as sphingomyelinase (SMase), have been associated with diarrheal syndrome [2]. The current study reports a draft genome sequence of a putative human pathogen *Bacillus* sp. strain SD-4 isolated from a cattle feed.

2. Materials and methods

The strain SD-4 was isolated from a cattle feed sample (33.54569, 71.45979) and was initially evaluated for its cytotoxicity and antibacterial activities against a set of ATCC indicator strains. Toxicity was evaluated by trypan blue dye assay, non-hemolytic toxin, and hemolysin activity were conducted with Duopath cereus enterotoxin (Merck) assay, according to the manufacturer's instructions. Motility test was performed using semisolid tryptic soy agar, and *Bacillus subtilis* RS10 [3] was used as a positive control. Biofilm formation was assessed on a microtiter plate by crystal violet staining. Antibacterial activity was determined using the soft agar overlay method as described earlier [3]. Antimicrobial susceptibility test was performed according to Clinical Laboratory Standard Institute (CLSI) protocols.

Total genomic DNA (gDNA) was extracted from fresh culture using a pure link gDNA extraction kit (Invitrogen, Carlsbad, CA, USA) according to the manufacturer's instructions. Whole-genome sequencing was performed using the Illumine Hiseq 2500 platform with paired-end reads (University of Birmingham, UK). The reads were assembled and the gaps between contigs were filled using GAPPadder [4]. The genome was annotated using NCBI Prokaryotic Genome Annotation Pipeline (PGAP) version 5.3, and secondary metabolites were identified using antiSMASH (https://antismash.secondarymetabolites.org/). Phylogenomic analyses were performed using the type strain genome server (TYGS) (https://tygs.dsmz.de/). MobileElement finder version 1.0.3 (https://cge.cbs.dtu.dk/services/MobileElementFinder/) was used to identify mobile genetic elements, and plasmids were predicted using PlasmidFinder version 2.1 (https://cge.cbs.dtu.dk/services/PlasmidFinder/). Virulence genes were determined using VirulenceFinder version 2.0.3 (https://cge.cbs.dtu.dk/services/VirulenceFinder/). Antibiotic resistance genes were identified by the comprehensive antibiotics resistance database (CARD) (https://card.mcmaster.ca). The draft genome sequence of *Bacillus* sp. strain SD-4 is deposited at Genbank under Bioproject PRJNA594265, Biosample SAMN21509075, and genome accession number CP083987.

3. Results and discussion

The genome of SD-4 was sequenced to 30x coverage and consists of 5 664 419 bps with a GC content of 35.18%. The genome carries 5328 coding genes, being 96 for tRNA, 13 for rRNA, five for ncRNAs, two plasmids, and five prophage regions. To determine the taxonomic placement of strain SD-4, average nucleotide identities (ANI) were calculated with the closely related strains *Bacillus paranthracis* and *Bacillus nitratireducens.* The genome of SD-4 strain showed a similarity of 95.43% (55.99% coverage) and 89.99% (52.14% coverage) to those strains, respectively. In-silico DNA-DNA hybridization (DDH) with *B. paranthracis* and *B. nitratireducens* were found to be 65.60% and 58.20%, respectively. The species cutoff values for ANI (96%) and DDH (70%) were used as previously proposed [5]. Furthermore, the whole genome-based phylogenomic tree of strain SD-4 and closely related strains shows the monoclade position of strain SD-4 with *B. paranthracis*, suggesting a novel candidate species (Fig. 1). Similarly, the type strain genome server (TYGS) also predicted that the strain SD-4 is a novel species as it does not belong to any species found in the TYGS database. Together, these results showed that the strain SD-4 is potentially a new species in the *Bacillus cereus* group.

Genome mining revealed 267 genes in 12 biosynthetic gene clusters (BGCs) coding for various bioactive metabolites, including fengycin, bacillibactin, siderophore, terpene, linear azol peptide (LAP), and thurincin H (Supplementary Table S1). These metabolites are produced by strain SD-4, which makes the strain capable of inhibiting the growth of ATCC indicator strains in the laboratory (Supplementary Fig. S1). In vitro motility and biofilm formation were confirmed by dispersing growth in semisolid agar and adhering of the strain SD-4 to the wall of the plate, respectively. The immunoassay showed no hemolysin activity, but non-hemolytic enterotoxin activity was detected. The in-silico analysis identified five antimicrobial resistance genes (ARGs), including *bla2* (carbapenem, cephalosporine resistance gene), *vanR* (glycopeptide resistance gene cluster), *fosB* (fosfomycin resistance gene), *mphL* (macrolides resistance gene), and *ant(4*́*)-lb* (aminoglycoside resistance gene) (Supplementary Table S2). However, results from the antimicrobial susceptibility tests showed that the strain is phenotypically resistant to glycopeptides (vancomycin 30 μg), β-lactams (ampicillin 25 μg), and aminoglycoside (streptomycin 25 μg) while susceptible to gentamycin (10 μg) and carbapenem (meropenem 10 μg) (Supplementary Fig. S2). The strain SD-4 genome contains two plasmid replicons, including repUS22 and rep20. The SD-4 genome carries at least four genomic islands, including *Listeria* LIPI-1 pathogenicity island. This island in SD-4 genome harbours genes encoding pathogenic proteins such as phosphatidylinositol-specific phospholipases C (*plcA*), thiol activated cytolysin (TACY), and broad substrate range phospholipases C (*plcB*).

Furthermore, 25 virulence genes were identified in the SD-4 genome (Supplementary Table S3). These genes are associated with iron regulation and acquisition, immune evasion, toxins, and type VII secretion system. The sphingomyelinase gene (*sph*) found in the SD-4 genome encodes a virulent factor involved in septicaemia. Additionally, two types of toxin-related genes, *cytK,* encoding cytotoxin K (hemolysin IV), and *nheC,* encoding a non-hemolytic enterotoxin, were identified (Supplementary Table S3). Based on the presence of gene families involved in pathogenicity, strain SD-4 was shown to have a 0.791 probability of being a human pathogen. Two intact prophages with sizes of 35.3 kbp and 48.5 kbp, and three incomplete phages were found in the genome.

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Declaration of Competing Interest

The authors declared no conflict of interest.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.jgar.2022.04.002.

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